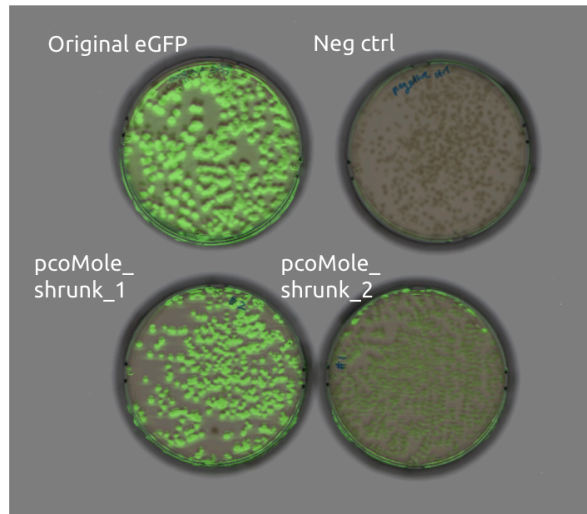


pCoMole wet-lab GFP results:

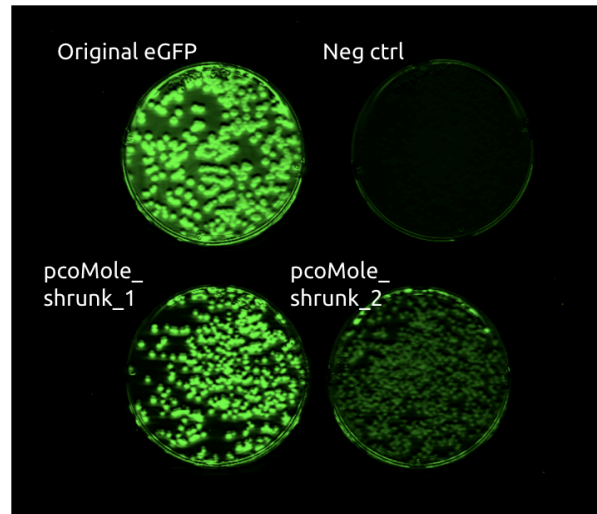
Method description:

We used pCoMole to shrink the eGFP sequence and tested the shrink proteins in BL21 cells for expression validation. The experimental results showed that two of the shrink variants retained green fluorescence. One variant contained one substitution and 10 deletions, while the other contained two substitutions and 10 deletions.

Phase contrast | eGFP (488 nm) | Merge



eGFP (488 nm)



Sequence details:

Original eGFP sequence: 239AA

```
MVSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLLKFICTTGKLPVPWPTLVTTLTYG  
VQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELKGIDFKEDGNILG  
HKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSA  
LSKDPNEKRDHMLLEFVTAAGITLGMDELYK
```

(more details on next page...)

Negative control sequence: 228AA, with **2 substitutions and 11 deletions**

MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTCLKFICTTGKLPVPWPTLVTTLTYGVC
FSRYPDHMKQHDFFKSAMPEGYVQERTISFKDDGNYKTRAEVKFEGDTLVNRIELKGIDFKEDGNILGH
KLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDYLSTSSPEKRD
HLEFVTAAGITLGMDELYK

Original eGFP	1	MVSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTCLKFICTTGKLPVPWPT	60
Negative ctrl	1	M-SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTCLKFICTTGKLPVPWPT	59
Original eGFP	61	LVTTLTYGVCFSRYPDHMKQHDFFKSAMPEGYVQERTISFKDDGNYKTRAEVKFEGDTL	120
Negative ctrl	60	LVTTLTYGVCFSRYPDHMKQHDFFKSAMPEGYVQERTISFKDDGNYKTRAEVKFEGDTL	119
*			
Original eGFP	121	VNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA	180
Negative ctrl	120	VNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLA	179
Original eGFP	181	DHYQQNTPIGDGPVLLPDNHYLSTQSALS KDPNEKRDHMLLEFVTAAGITLGMDELYK	239
Negative ctrl	180	DHYQQNTPIGDGPVLLPD--YLSTSS-----P-EKRDH--LLEFVTAAGITLGMDELYK	228
*			